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| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 07/23/2011 |
| **PI Name:** | Pluth, Janice Ph.D. | | |
| **Project Title:** | Elucidating the Relationship Between the Effects of Various Radiation Qualities and Cancer Development Processes using Novel Flow-based Assays | | |
| **Division Name:** | Human Research | | |
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| **Zip Code:** | 89154-9900 | **Congressional District:** | 1 |
| **Comments:** | NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017. | | |
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| **Key Personnel Changes/Previous PI:** | | | |
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| **Task Description:** | Current radiation quality factors used for determining cancer risk for space radiation do not provide the accuracy needed, and there is debate as to the relevance of the experimental studies they're based upon. Therefore, one of NASA's primary goals is to improve the understanding of the carcinogenic process related to high LET exposures to better define the risk. Recently our current understanding of carcinogenesis has improved, and a number of cancer hallmarks have been identified. Some hallmarks, or surrogate cancer endpoints include genetic instability, telomere length changes and aberrant cellular signaling. Identifying relationships between early and late cell signaling profiles, later surrogate cancer endpoints, and radiation qualities will aid in determining a better risk assessment for cancer formation. In addition, using relevant cell types, such as human epithelial cells in which the majority of cancers develop is also key. In this proposal we will study critical phosphorylation signaling at early and late time points as well as surrogate cancer endpoints of telomere length changes and genomic stability as measured by centrosome amplification, at these same late times post exposure. Epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and why certain cell types more easily become carcinogenic. We will also perform studies to understand the heterogeneous nature of high LET radiation. Energy deposition from high LET ions is dense in the track cores, resulting in closely spaced and complex damages, and delta rays originating from cores provide additional exposures similar to X-rays. Using carefully chosen charge, energy and LETs of particles to obtain differing populations of cells hit by tracks and delta rays alone we will investigate how phosphorylation profiles change. These studies will allow us to better quantify the effects of space radiation and its relationship to the carcinogenic process. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | In this proposal we are studying critical phosphorylation signaling at early (< 24 h) and late time points (days-weeks) following exposure. In addition, we will determine the relationship between these phospho-protein patterns, and two indicators of genomic stability and cancer progression, telomere shortening and centrosome amplification. Breast cancer studies have revealed that telomere length decreases during progression from usual ductal hyperplasia (UDH) to invasive cancer, thus telomere length changes can provide a readout of the carcinogenic process. A second marker we will be following after exposure is centrosome amplification which involves gains and losses of whole chromosomes and is believed to be due to continuous chromosome mis-segregation during mitosis resulting from centrosome defects. Combinations of increases in number, size, abnormal structure and function characterize centrosome amplification. Centrosome amplification has been detected in a broad range of solid tumors and more recently in leukemia and lymphoma. Centrosome abnormalities have been shown to increase with increasing tumor grade and aggressiveness. Over-expression of pericentrin, a centrosome protein, mimics many of the cellular characteristics of a high-grade carcinoma. Therefore centrosome abnormalities are also a very good indicator of cancer progression. Both of these potential markers for carcinogenesis will be followed using flow based assays.<br>A difficulty with high-LET exposures, particularly at low doses, is that very uneven levels of damage are deposited within the population of cells. When studying effects at longer times post low dose exposure healthy cells will outgrow the population making it difficult to study effects in the minor population more severely affected. To study populations of cells exhibiting varying levels of damage we will isolate and individually expand populations separately for analysis of surrogate carcinogenic endpoints. We will utilize a reagent called CFSE which allows one to identify cells blocked at various cell cycles following damage, and thus isolate more heavily damaged cells based on cell cycle delays. Sets of mammary epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and whether certain cell types more frequently show evidence of genetic instability upon radiation exposure. These studies will complement our earlier studies using a normal hTERT immortalized skin fibroblast line, and help us identify unique signaling patterns specific to different cell types. By using two cell types from the same individual we will identify genotype influences in response. The relevance of considering d-rays when determining the risk of high-LET radiation will also be studied. Carefully chosen particles of various energies will be used to obtain differing populations of cells hit in a greater proportion by tracks or d-rays to investigate how phosphorylation profiles and carcinogenic endpoints are affected by these components. The intent of this proposal is to provide a comprehensive picture of HZE effects from initial phosphorylation patterns to later phosphorylation profiles, and the relationships to various surrogate markers for carcinogenesis. Studies will be performed using novel flow cytometry-based assays, which have been optimized in our lab, to more rapidly and less subjectively monitor these endpoints.<br><br>Materials & Methods: Cells were set up in flasks or chamber slides prior to irradiation, irradiated at the various doses and then fixed at multiple time points thereafter. A methanol and a combination paraformaldehyde and methanol fixation were used. Fixed slides were sent back to our home institution in PBS to stain with appropriate primary and secondary antibodies. Cells in flasks were trypsinized off and fixed, then sent back on ice in methanol. Aliquots of fixed cells are then stained for various phospho-proteins and analyzed via flow cytometry. For longer timepoint analysis (weeks post exposure) cells are grown to confluence while at Brookhaven labs then frozen down before leaving (~ 1 week post exposure). Cells are then thawed back at the home institution and grown to confluence again to then analyze for telomere length changes and centrosome defects. Cells are fixed with a methanol fix for centrosome analysis by flow and are counted and mixed with a hybridization buffer containing telomere probe while heating to allow access of probe to telomeric DNA.<br><br>Brief summary of past years research accomplishments: During the first year efforts were concentrated on obtaining the phospho-protein signature in a normal fibroblast line (82-6) following exposure to various combinations of ions selected to best identify the contribution of the delta-rays versus the direct track biological effects. Subsequent years (2-3) were focused on obtaining samples in triplicate for both this line and isogenic mammary fibroblast and epithelial strains. Difficulties in staining mammary samples were noted and eventually overcome in the third year of this project. We are currently completing analysis of mammary samples, and have completed all skin fibroblast analysis. We now have a full detailing of kinetics for 10 different energy/ion combinations for skin fibroblast cells. We have noted various temporal phospho-protein dependent differences. At early times post exposure Ti326 reveals the greatest induction of DSBs, whereas at late timeponts Fe 300, Fe 600 and Ti326 are nearly equivalent. As residual breaks (24h post) are linked to an increased cancer risk this timepoint is likely more biologically relevant. In contrast, for the two ATM mediated phosphorylations we see some difference from each other as well as from gH2AX kinetics. Both reveal major induction following Ti 326 exposure, however at 24 h post exposure pSMC1 is highly elevated following Ti 326, whereas pATF2 shows maximal residual levels following Si 173 and Fe 600. Single-cell flow cytometry data was exported to our collaborators at Johnson Space Center (JSC) to now analyze to define subtler differences perhaps missed when data is averaged over the total population. We will wait to make final conclusions regarding whether phospho-signaling kinetics can be described based on different radiation quality descriptions until analysis on the single cell basis has been |

completed.

Additional studies have been completed for surrogate cancer endpoints of telomere length and centrosome defects at many days to weeks post exposure in the same set of cell strains and lines. Results were exported from FACS files so that individual cells could be analyzed for individual cell telomere length changes and centrosome defects. Some preliminary results have been completed but during this process we have noted that some changes needed to be made in the experimental set up and thus during the last Brookhaven run we exposed samples in triplicate for both telomere and centrosome assay endpoints to have additional samples to compare using this new optimized protocol. We also have noted during analysis that at least two populations appear to be present in primary mammary cells and the numbers of cells in each population tend to vary dependent on dose and culture time post exposure. We have initiated studies to more carefully define these cells and believe they may be a population that is becoming senescent due to radiation exposure and continued culture. The smaller (low FSC) and less granular (low SSC) population appears to contain the cells that are primarily dividing and thus may be the population to concentrate on for analysis. We are currently in the process of thawing and growing all the new centromere and telomere samples from the last run and will process and stain those in the final year of this project to then send for single cell analysis at JSC.

Preliminary flow-based centrosome studies have been completed on a single cell basis for three independent Si 93 MeV/u exposed mammary epithelial cell experiments. Samples were stained to allow both centrosome defects and DSBs (gH2AX) to be defined. We observe a non-linear, dose-dependent increase in centrosome aberrations. However considerable inter-experimental variation appears evident which may have resulted from differing growth time following exposure. Thus with subsequent experiments all growth time is being held consistent and any experimental variables that we can confine will be. Roughly 10-30% of the dosed cells showed higher levels of centrosome aberrations as compared to controls, with lower doses (0.05 Gy) showing less (10%) than the higher 1 Gy dose (30%). All centrosome analysis must be performed on single cell basis as this is a somewhat rare event and would be lost if averaged data was used. When viewing the relationship between persistent gH2AX foci and centrosome levels we have noted a significant correlation between increased gH2AX and higher centrosome aberrations (Person’s p of 0.247, 95%=0.188 10 0.295) when analyzing the total population of cells. In addition, with flow cytometery data you can specifically gate on cells with either only high pericentrin, both high pericentrin and high gH2AX, or high gH2AX only. When this was done and cells showing both high levels of gH2AX and high levels of pericentrin were graphed we were then able to observe a more linear response with dose. These studies are of interest in that very low doses (0.05 Gy) appear to induce both centrosome defects and persistent gH2AX foci (5-10% of cells) and genomic instability and persistent foci are known to correlate highly with malignant progression (Bartkova, 2005). In this final year of the project we plan to grow, fix and stain the samples recently obtained in triplicate for each ion/energy for both mammary fibroblast and epithelial cells to then compare centrosome and gH2AX responses between tissue type and radiation quality.

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| **Bibliography Type:** | Description: (Last Updated: 11/01/2023) |
| **Articles in Peer-reviewed Journals** | Chappell LJ, Whalen MK, Gurai S, Ponomarev A, Cucinotta FA, Pluth JM. "Analysis of flow cytometry DNA damage response protein activation kinetics after exposure to x rays and high-energy iron nuclei." Radiat Res. 2010 Dec;174(6):691-702. Epub 2010 Sep 28. PMID: 21128792 , Dec-2010 |